Competitive Coding

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Experiment - 7

Approach :-

(a) Take cities as graph nodes and flights as edges with cost.

(b) Use BFS for traversal
where each level = one more stop

(c) We will keep track of :-
- cost (minimum price)
- stops (constraint)

Pseudocode :-

```
push (src, 0 cost, 0 stops)

dist[] = [∞, ∞, ... ∞]
dist[src] = 0

while q not empty :
        (node, cost, stops) = q.pop()

        if stops > k :
                continue

        for each (neighbour, price) in adj[node] :

                if cost + price < dist[neighbour] :
                        dist[neighbour] = cost + price
                        push (neighbour, dist[neighbour],
                                stops + 1)
```

```
if dist[dst] == ∞ :
        return -1
else
        return dist[dst]
```

<u>Time Complexity</u> :-

$O(E \times K)$

<u>Space Complexity</u> :-

$O(E+V)$

$E \rightarrow$ number of flights

$K \rightarrow$ number of stops